

PRISCILA CANO

Cropping Systems | Digital Agriculture | Sustainable Intensification

West Lafayette, IN, 47906 | 217 607 4370 | priscilacano8@gmail.com

Research Program

My research integrates cropping systems, environmental characterization, and digital agriculture tools to improve resource-use efficiency and crop performance under variable environmental conditions. My work combines field experimentation, spatial data analysis, and producer decision-making research to develop sustainable, data-driven, and practical crop management strategies.

My research integrates cropping systems, environmental characterization, and digital agriculture tools to improve crop performance and support practical, data-driven decision-making in agricultural systems.

EDUCATION

Ph.D. Agricultural Sciences , University of Buenos Aires, Argentina	October 2023
Dissertation: Sustainable intensification in crop rotation	
Principal Advisor: Dr. Santiago Poggio	
Spec. Remote Sensing , University of Buenos Aires, Argentina	October 2020
Dissertation: Impacts of irrigation-driven land-use change on vegetation functioning in semi-arid systems	
Principal Advisor: Dr. María Eugenia Beget	
B.S. Agricultural Sciences , UNNOBA, Buenos Aires, Argentina	February 2017
Dissertation: Nutrient balance-based evaluation of environmental costs in agricultural systems	
Principal Advisor: Dr. Silvina Cabrini	

RESEARCH INTERESTS

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- Cropping systems
 - Field experimentation
 - Crop rotations and cover crops
 - Sustainable intensification
 - Digital agriculture
 - Remote sensing
 - Spatial data analysis
 - Farmer decision-making
 - Agronomic systems analysis

SKILLS

- Remote sensing and spatial analysis (satellite and UAV imagery; Google Earth Engine, ArcGIS, QGIS)
- Statistical and data analysis (R, Python)
- Language Spanish (native) - English (proficient)
- Driver's license

RESEARCH APPOINTMENTS

February 2025 **Purdue University, Department of Agronomy - PI: Ignacio Ciampitti**
 Present Postdoctoral Researcher – Cropping Systems & Digital Agriculture

- Leading interdisciplinary research integrating agronomy, remote sensing, and data analytics to understand environmental drivers of maize yield variability across the U.S. Midwest.
- Conducting envirotyping analyses linking climate and soil conditions with crop productivity.
- Designing and implementing multi-state surveys on farmer and retailers adoption of digital agriculture technologies.
- Developing spectral approaches using drone and satellite imagery to assess crop nutrient status.

March 2024 **Kansas State University, Department of Agronomy - PI: Ignacio Ciampitti**
 January 2025 Postdoctoral Researcher

- Led a comprehensive review of U.S. agricultural technologies across nutrient management, sensing, and decision-support systems.
- Conducted a review of water use efficiency in sorghum and corn in Kansas
- Designed and guided field soybean sampling protocols, and oversaw sample collection to support nutrient conservation and farmer-oriented decision-support tools.

August 2023 **University of Illinois at Urbana-Champaign, Department of Agricultural**
 February 2024 **and Consumer Economics – David Bullock**
 Visiting Scholar

- Participated in large-scale on-farm trials with the DIFM (Data Intensive Farm Management) group, deploying precision agriculture technologies to support decision-making.
- Processed and analyzed high-resolution spatial data to inform optimized nitrogen management strategies.

January 2015 **National Institute of Agricultural Technology, Argentina**
 January 2024 Researcher

- Led a 5-year on-farm experiment evaluating crop rotations and sustainable intensification strategies in the Argentine Pampas.

- Designed and implemented field experiments comparing soybean monoculture, maize–soybean rotations, and diversified wheat-based systems with cover crops.
- Quantified agronomic, economic, and environmental trade-offs using multi-criteria decision analysis.
- Results published in **Agricultural Systems (2023)**.

PEER REVIEWED JOURNAL PUBLICATIONS

- **Cano, P.B.**, Madonni, G.A., Gomez, F., Mandrini, G., Palmero., F. (in preparation) Yield-driven environmental characterization of rainfed maize production regions in Indiana
- **Cano, P.B.**, Mandrini, G., Erickson B., Buckmaster D., Arnall, B., Ajay, S., Balboa, G., Carcedo, A., Hawkins, E., Fulton, J., Froes de Borja Reis, A., Kent, S., Ken, S., Kovac, P., Steve, P. Zhou, J. and I. Ciampitti (2026) Farmer Perspectives on Digital Agriculture in the US Midwest. *Agronomy Journal*, 118(1), e70268.
- **Cano, P.B.**, Cabrini, S.M., & Poggio, S.L. (2025). Decision-making on crop sequence diversification in the Argentine pampas. *Land Use Policy*, 159, 107778.
- **Cano, P.B.**, Carcedo, A.J., Hernandez, C.M., Gomez, F.M., Gimenez, V.D., Kyveryga, P.M., & Ciampitti, I.A. (2025). Trends in agricultural technology: a review of US patents. *Precision Agriculture*, 26(4), 59.
- **Cano, P. B.**, Cabrini, S.M., Peper, A.M., & Poggio, S.L. (2023). Multi-criteria assessment of cropping systems for sustainable intensification in the Pampas. *Agricultural Systems*, 210, 103723.
- Cabrini, S.M., Portela, S.I., **Cano, P.B.**, & López, D.A. (2019). Heterogeneity in agricultural land use decisions in Argentine Rolling Pampas: The effects on environmental and economic indicators. *Cogent Environmental Science*, 5(1), 1667709

BOOKS

- Kyveryga, P., **Cano, P.**, Cisdeli, P., Hernández, C., Santiago, G., and Ciampitti, I. (2024). Developments in modelling and decision support systems in precision agriculture and conservation. in *Precision Agriculture for Sustainability: Second edition* (ed. John Stafford) (accepted).
- Ciampitti, I., Mandrini, G., Gomez, F., Cisdeli, P., Nocera, G., **Cano, P.**, Lacasa, J., Palmero, F., Cammarano, D., & Kyveryga, P. (2025). Digital agriculture to transform decision support systems: Current state and perspectives. In D. L. Sparks (Ed.), *Advances in Agronomy*. Elsevier.

EXTENSION

VI Seminar on Agricultural Risk Management (2025)

National Institute of Agricultural Technology (INTA) – Center for Research in Economics and Foresight (CIEP)-Universidad de Buenos Aires (UBA)

Invited Panelist “Digitalization, AI, and Decision-Making in Agriculture”

- Participated as an invited panelist discussing the role of digital technologies and artificial intelligence in agricultural risk management and decision-making.

KARA Summer Field School, Kansas State University (2024)

- Participated in intensive in-service training alongside Extension agents, agronomists, and crop consultants, focusing on corn and sorghum growth, development, and production challenges.
- Gained hands-on experience in translating research findings to practical recommendations for growers.

Soybean Project Extension Support, Kansas State University (March 2024 – January 2025)

- Contributed to extension materials and farmer-facing reports derived from a multi-year, multi-state database.

Field days – Cropping Systems Trials, INTA, Argentina (2015–2021)

- Organized and led on-farm field days showcasing the development and results of long-term cropping systems trials.
- Engaged ~30 producers per event, presenting agronomic practices, sustainability indicators, and preliminary results.
- Facilitated interactive discussions to gather farmer feedback and encourage adoption of improved crop sequences and conservation practices.

Chacras group-APRESID Crop sequence evaluation with farmers, INTA, Argentina (2018)

- Worked directly with producers to evaluate different crop sequences, integrating farmer knowledge and practical constraints.
- Monitored field trials and conducted farmer interviews to evaluate agronomic performance, sustainability indicators, and adoption potential of different crop rotations.

Organic vs. conventional production evaluation, INTA, Argentina (2017)

- Collaborated with a network of organic producers to evaluate sustainability indicators compared to conventional cropping systems.
- Conducted farm-level assessments and generated recommendations to improve productivity while minimizing environmental impact.

TECHNICAL PUBLICATIONS

- Cano, P.B.**, Erickson, B., Lowenberg-Deboer, J., & Ciampitti I. (with editors). The promise and reality of AI and autonomous technology in ag retail. *Crop Life*.
- Cano, P. B.**, & Ciampitti, I. A. (2025). Digital Technologies: Mapping the Future of Corn Production and US Agriculture. *Indiana Corn Research and Update, Issue #24 – October 6, 2025*.
- Cano, P.**, Cabrini, S., & Poggio, S. (2023). What crop sequences predominate in Pergamino? *Revista RTA INTA*.
- Gatti, N., Amaro, I. B., & **Cano, P. B.** (2023). Caminante, no hay camino, se hace camino al andar: experiencias de administración de caminos rurales en la provincia de Buenos Aires. *Revista Argentina de Economía Agraria*, 24(1), 52–63.
- Cano, P.**, Cabrini, S., & Portela, S. (2017). Environmental cost associated with nitrogen and phosphorus balances in agricultural enterprises in Pergamino. *Revista de Tecnología Agropecuaria – INTA*, 10, 42–45.

CONFERENCE PRESENTATIONS

- **Cano, P.B.** (2025) Farmer Perspectives on Digital Agriculture in the U.S. Midwest. ASA, CSSA, SSSA International Annual Meeting. Salt Lake City, Utah, United States. (talk)
- **Cano, P.B.** (2025) Yield-driven environmental characterization of rainfed maize production regions in Indiana. First Purdue Symposium of Digital Agriculture (SyDag 2025) Purdue University, West Lafayette, Indiana, USA (poster)
- Mangidis, V., Cano, P., & Ciampitti, I. A. (2026). *Predicting wheat quality using satellite data*. First Purdue Symposium of Digital Agriculture (SyDag 2025) Purdue University, West Lafayette, Indiana, USA (poster)
- **Cano, P.B.** (2024) Trends in Agricultural Technology: A Review of US Patents. ASA, CSSA, SSSA International Annual Meeting. San Antonio, Texas, United States. (talk)
- Parola, P., Gómez, F., Carcedo, A. J., **Cano, P. B.**, & Ciampitti, I. (2024). *Influence of observed versus gridded soil texture and organic carbon on simulated soil N mineralization and water content: A case study in Senegal* [Abstract]. ASA, CSSA, SSSA International Annual Meeting, San Antonio, TX, USA. (poster)
- **Cano, P.B.** (2024) Trends in Agricultural Technology: A Review of US Patents. 16th International Conference on Precision Agriculture in Manhattan, KS.(talk)
- **Cano, P.B.** (2023) Towards Sustainable Intensification: Decision-Making on Crop Sequence Diversification in the Pampas Region. IAAE Interconference Symposium. Montevideo, Uruguay.(talk)
- **Cano, P.B.** (2020) Economic and Environmental Evaluation of Alternatives to Intensify and Diversify Agricultural Rotations in Buenos Aires Province. Agrarian Economy Congress. Pergamino, Argentina. (talk)
- **Cano, P.B.** (2019) Economic Evaluation of Alternatives to Intensify and Diversify Agricultural Rotations in Northern Buenos Aires Province. Annual Meeting of Agricultural Economics. Faculty of Agronomy, University of Buenos Aires. Buenos Aires, Argentina.(talk)

TEACHING

- Teaching Assistant – **AGR 59500: Research Fundamentals in the Era of AI Purdue University** | March 2026 – April 2026
 -Supported course instruction focused on research design, data analysis, and the integration of AI tools in agricultural research.
 -Assisted students with assignments and project development, providing guidance on data-driven approaches and analytical methods.
 -Facilitated discussions on the application of emerging technologies in agronomy and agricultural systems research.
 -Supported student learning in data analysis, research design, and the application of digital tools in agronomy.
- Teaching Assistant – **Agricultural Economics**
 National University of the Northwest of Buenos Aires, Argentina | Aug 2013 – Dec 2019
 Under the supervision of Prof. Dra. Silvina Cabrini
 -Managed multiple-choice exams, ensuring accurate grading and assessment integrity.
 -Led consultation sessions for students, providing guidance on coursework and problem-solving.

-Assisted in practical classes, supporting hands-on learning and reinforcing theoretical concepts

- **Teaching Assistant – Experimental Design**

National University of the Northwest of Buenos Aires, Argentina | Aug 2013 – Dec 2015

Under the supervision of Prof. Dra. Griselda Negri

-Developed practical guides for evaluating different field experimental design methodologies.

-Assisted in practical classes, supporting students in implementing experimental designs in the field.

-Collaborated with instructors to facilitate hands-on learning and ensure smooth class execution.

MENTORING STUDENT

- Mentored Natalia Volpato (Ph.D. student) on cover crop evaluation, including experimental design, data analysis, and interpretation for her doctoral thesis. (2025-present)
- Supported Paulina Parola (Visiting Scholar) in research on soil processes, including data processing, analysis, and interpretation of results related to soil texture, organic carbon, and nitrogen mineralization. (2024)

CERTIFICATIONS

- Agricultural Estimations and Earth Observation Technologies
University of Maryland, Department of Geographical Sciences | October 2022

HONORS

- **Finalist – “Ing. Agr. Gerardo Bartolomé” Award (2024)**
Graduate School Alberto Soriano & Grupo Don Mario, Faculty of Agronomy, University of Buenos Aires
Recognized among the top three doctoral theses in the 2024 Best Postgraduate Thesis in Extensive Crops competition.
- **Second Honorable Mention "Enrique Chaneton" Award (2021)**
Graduate School "Alberto Soriano," Faculty of Agronomy, University of Buenos Aires
Recognize graduate students who present their research in a clear, accessible manner and highlight its societal impact.
- **Best Paper Mention – 53rd Annual Meeting of the Argentine Association of Agricultural Economics (2022)**
Argentine Association of Agricultural Economics
Recognized for the research paper: "Decision-Making on Crop Sequence Diversification in the Pampas Region".

GRANTS

- **USDA / North Central Soybean Research Program (NCSRP) (2022–2024)**
Project: Mapping Soybean Protein and Oil Quality in Farmer Fields
Funding: USD 218,000
PI: Ignacio Ciampitti (Kansas State University)
Role: Research collaborator
- Collaborated in a multi-state (10 states) soybean quality project integrating field sampling, yield data, and remote sensing.
- Contributed to the development of regional databases of soybean protein and oil concentration across farmer fields.
Supported data synthesis, reporting, and preparation of extension-oriented outputs for farmers and stakeholders.
- **National University of the Northwest of Buenos Aires (UNNOBA) (2019–2021)**
Project: Economic and Environmental Evaluation of Sustainable Intensification Alternatives in the Arrecifes River Basin
Project Code: 0611/2019
PI: Silvina M. Cabrini (UNNOBA)
Co-PI: Santiago L. Poggio (University of Buenos Aires)
Role: Investigator

- Contributed to the economic and environmental assessment of alternative cropping systems under sustainable intensification strategies at the watershed scale.
- Supported analyses of productivity, economic returns, environmental indicators, and trade-offs among management alternatives in the Arrecifes River Basin.
- **University of the Northwest of Buenos Aires (UNNOBA) – Monsanto Argentina (2019 –2021)**
Project: Economic and environmental evaluation of intensification and diversification alternatives in grain production systems in the Argentine Pampas
Funding: USD 3,000
PI: Silvina M. Cabrini (UNNOBA)
Industry Partner: Monsanto Argentina S.R.L.
Role: Investigator (Doctoral Fellow, CIC)
- Participated in an industry-sponsored research agreement evaluating economic performance and eco-efficiency of grain production systems under different levels of agricultural intensification and diversification.
- Contributed to the calculation of economic indicators (gross margins, return rates, risk) and environmental indicators, including agrochemical environmental impact (EIQ), water balance, nutrient balances, and eco-efficiency metrics.
- Supported the preparation of technical reports and analyses used for industry-oriented dissemination and decision support.
- **National University of the Northwest of Buenos Aires (UNNOBA) (2017–2019)**
Project: Dynamics, sustainability, and value-added processes of agricultural production systems in northern Buenos Aires Province
Funding: ARS 50,000
PI (Director): Carlos Senigaglia (UNNOBA)
Co-PI (Co-director): Silvina M. Cabrini (UNNOBA)

Role: Investigator

- Contributed to a research project identifying challenges and opportunities for agricultural production systems in northern Buenos Aires, focusing on sustainability, economic performance, and value-adding processes.
- Supported analyses of land-use change, land tenure dynamics, and economic and environmental performance of extensive cropping systems, using Pergamino County as a case study.
- Contributed to evidence generation to inform strategies and policies supporting small producers, diversification, and territorial sustainable development.

FELLOWSHIPS & SCHOLARSHIPS

Scientific Research Commission of Buenos Aires Province (CIC) (2017 - 2019)

Doctoral Fellowship

Host Institution: National Institute of Agricultural Technology (INTA), Pergamino Experimental Station

Funding: 100% doctoral funding

Advisor (PI): Silvina M. Cabrini

World Bank-funded Project (BIRF P176905) (2023 - 2024)

Research Fellowship – International Training Program

- Awarded a competitive research and training fellowship under INTA's Graduate and Professional Development Program to support an international research stay.
- Fellowship funded through the World Bank (IBRD) project "Inclusive and Climate-Smart Agri-food Systems" (P176905), covering international travel and living expenses.

ACADEMIC SERVICES

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| • Reviewer Agronomy Journal | December 2025 |
| • Committee Member Post Doc Association | August 2025-present |
| • Purdue Agronomy Graduate Student Organization (Role: mentor) | August 2025-present |
| • American Society of Agronomy, Crop Science, and Soil Science Member | October 2024-present |
| • International Society of Precision Agriculture Member | January 2024-present |